

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
Student Name:	R. V. Harsha Vardhan	Reg. No.:	192211868
Section / Batch:	CSE/2022	Date:	22/07/2026

Code of Conduct

I R. V. Harsha Vardhan(192211868) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: R.V.Harsha vardhan

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25

	GRAND TOTAL:	100 Marks
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Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge -case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, welldocumented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and welljustified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

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Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List Data Structures

5 marks

Q9

Linked List

5 marks

Q10

Queue

5 marks

if(arr[mid] == key)
return mid;
else if(arr[mid] < key)
low = mid;
else
high = mid;
}
return -1;
}

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarySearch(int arr[], int n, int key) {  
2     int low = 0, high = n - 1;  
3     while (low < high) {  
4         int mid = low + (high - low) / 2;  
5         if (arr[mid] == key)  
6             return mid;  
7         else if (arr[mid] < key)  
8             low = mid + 1;  
9         else  
10            high = mid - 1;  
11     }  
12     return - 1;  
13 }
```

Test Input

stdin input (optional)

Output

Compilation Error:
C:/OCPPComp/OCPP2/bin/./lib/gcc/
x86_64-w64-
mingw32/10.3.0/./.././../x86_64-
w64-mingw32/bin/ld.exe:
C:/OCPPComp/OCPP2/bin/./lib/gcc/

Pending manual review

10/10 answered



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Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List Data Structures

5 marks

Q9

Linked List

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {  
    if(top == 0) {  
        printf("Underflow");  
        return -1;  
    }  
    return stack[top--];  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int pop() {  
2     if (top == 0) {  
3         printf("Underflow");  
4         return -1;  
5     }  
6     return stack[top--];  
7 }
```

Test Input

stdin input (optional)

Output

▶ Run

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🚀 Submit Fix

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Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List Data Structures

5 marks

Q9

Linked List

5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int stack[5];
2 int top;
3
4 void init()
5 {
6     top = -1;
7 }
```

Test Input

stdin input (optional)

Output

▶ Run

💾 Save

✳ Submit Fix

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Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List Data Structures

5 marks

Q9

Linked List

5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {
while(stack[top] != ')') {
postfix[j++] = pop();
}
pop();
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if (ch == ')') {
2     while (stack[top] != '()') {
3         postfix[j++] = pop();
4     }
5     pop();
6 }
```

Test Input

stdin input (optional)

Output

Compilation Error:
C:\Windows\TEMP\sandbox_192211068RB_6a8be9bed92dc8.00915861\main.c:11:1: error: expected identifier or '(' before 'if'
1 | if (ch == ')') {
|

Run

Save

Submit Fix

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q5 Stack

5 marks

In balancing Bracket, you identify the error.

```
if((ch == '[' && stack[top] == '[') ||
(ch == ']' && stack[top] == '[')) {
    pop();
} else {
    return 0;
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if (top == -1) {
2     return 0;
3 }
4
5 if ((ch == '[' && stack[top] == '[') ||
6     (ch == ']' && stack[top] == '[') ||
7     (ch == '[' && stack[top] == '[')) {
8     pop();
9 } else {
10    return 0;
11 }
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_192211868
RB_6a82C1a4161800.65981360\main.c
:1:1: error: expected identifier
or '(' before 'if'
```

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Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List Data Structures

5 marks

Q9

Linked List

5 marks

What is the issue?

temp = head

while temp.next != NULL:

print(temp.data)

temp = temp.next

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 temp = head
2 while temp is not None:
3     print(temp.data)
4     temp = temp.next
```

Test Input

10 20 30

Output

Compilation Error:
C:\Windows\TEMP\sandbox_192211068
RB_6a82c2c50abb0.56639108\main.c
:1:1: warning: data definition
has no type or storage class
1:1 temp = head

Pending manual review

Submitted

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q7 Stack

5 marks

Point out the error.

POP(stack):

if top == 0:

print("Underflow")

else:

return stack[top--]

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 def POP(stack):
2 if top == -1:
3 print("Underflow")
4 else:
5 item = stack[top]
6 top -- 1
7 return item
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_192211868
RB_6a82c3f0b6e287.73163137\main.c
:1:1: error: unknown type name
'def'
1 1 def POP(stack):
```

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QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List Data Structures

5 marks

Identify the errors in the following code

while current != NULL:
current.next = prev
prev = current
current = current.next

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 while current != NULL:  
2 next = current.next  
3 current.next = prev  
4 prev = current  
5 current = next
```

Test Input

stdin input (optional)

Output

```
Compilation Error:  
C:\Windows\TEMP\sandbox_192211868  
RB_6a82c4a7b20900.239999999\main.c  
:1:1: error: expected identifier  
or '(' before 'while'  
1 while current != NULL:
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q9

Linked List

5 marks

What is the error in the below code?

new.next = head

head = new

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 new->next = head;
2 head = new;
```

Test Input

stdin input (optional)

Output

▶ Run

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q10 Queue

5 marks

What is missing in the below pseudocode?

DEQUEUE(Q):
if front == -1:
 print("Underflow")
else:
 return Q[front++]

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 DEQUEUE(Q):  
2 if front == -1:  
3   print("Underflow")  
4   else:  
5     item = Q[front]  
6   if front == rear:  
7     front = rear = -1  
8   else:  
9     front = front + 1  
10    return item
```

Test Input

stdin input (optional)

Output

```
Compilation Error:  
C:\Windows\TEMP\sandbox_192211868  
RB_6a8bc9fc869674.58706148\main.c  
:1:1: warning: return type  
defaults to 'int' [-Wimplicit-  
int]
```


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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack Notations - Postfix Expression

5 marks

✓

Q2

Circular Queue

5 marks

✓

Q3

Binary Search

5 marks

✓

Q4

Queue

5 marks

✓

Q5

Queue

5 marks

✓

Q6

Doubly Linked List

5 marks

✓

Q7

Stack

5 marks

✓

Q5 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void enqueue(int data) {  
2     if ( rear == MAX - 1)  
3         return;  
4  
5     queue[++rear] = data;  
6 }
```

Test Input

stdin input (optional)

Output

Compilation Error:

C:\Windows\TEMP\sandbox_192211868
RB_6a8bc790eccc44.54142576\main.c
: In function 'enqueue':
C:\Windows\TEMP\sandbox_192211868
RB_6a8bc790eccc44.54142576\main.c

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack Notations - Postfix Expression

5 marks

Q2

Circular Queue

5 marks

Q3

Binary Search

5 marks

Q4

Queue

5 marks

Q5

Queue

5 marks

Q6

Doubly Linked List

5 marks

Q7

Stack

5 marks

Q6 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {
    if (del->prev != NULL)
        del->prev->next = del->next;
    if (del->next != NULL)
        del->next->prev = del->prev;
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void deleteNode(Node* del){
2     if (del->prev != NULL)
3         del->prev->next = del->next;
4     else
5         head = del->next;
6
7     if (del->next != NULL)
8         del->next->prev = del->prev;
9 }
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_192211868
RB_6a8bc82fd1/a20.76905823\main.c
:1:17: error: unknown type name
'Node'
```


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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack Notations - Postfix Expression

5 marks

Q2

Circular Queue

5 marks

Q3

Binary Search

5 marks

Q4

Queue

5 marks

Q5

Queue

5 marks

Q6

Doubly Linked List

5 marks

Q7

Stack

5 marks

Q7 Stack

5 marks

Identify the issue in the below pseudocode.

POP():
if top == 0 → Underflow

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 pop():  
2 if top == -1 -> Underflow  
3 else:  
4 return stack[top--]
```

Test Input

stdin input (optional)

Output

```
Compilation Error:  
C:\Windows\TEMP\sandbox_192211060  
RB_6a8bc8747bd656.67850296\main.c  
:1:1: warning: return type  
defaults to 'int' [-Wimplicit-  
int]
```

▶ Run

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🚀 Submit Fix

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack Notations - Postfix Expression

5 marks

Q2

Circular Queue

5 marks

Q3

Binary Search

5 marks

Q4

Queue

5 marks

Q5

Queue

5 marks

Q6

Doubly Linked List

5 marks

Q7

Stack

5 marks

Q8 Stack

5 marks

Point out the error.

if top == MAX:

Overflow

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if top == MAX - 1
2     overflow
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_192211060
RB_6a8bc8a3effc02.52396646\main.c
:1:1: error: expected identifier
or '(' before 'if'
1 1 if top == MAX - 1
```

▶ Run

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack Notations - Postfix Expression

5 marks

Q2

Circular Queue

5 marks

Q3

Binary Search

5 marks

Q4

Queue

5 marks

Q5

Queue

5 marks

Q6

Doubly Linked List

5 marks

Q7

Stack

5 marks

Q9

Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
printf("Queue Full");
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if ((rear + 1) % MAX == front) {
2     printf("Queue Full");
3 }
4 else {
5     rear = (rear + 1) % MAX;
6     queue[rear] = vehicle;
7 }
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_192211868
RB_6a8bc907148537.85299140\main.c
:1:1: error: expected identifier
or '(' before 'if'
1 1 if ((rear + 1) % MAX ==
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack Notations - Postfix Expression

5 marks

Q2

Circular Queue

5 marks

Q3

Binary Search

5 marks

Q4

Queue

5 marks

Q5

Queue

5 marks

Q6

Doubly Linked List

5 marks

Q7

Stack

5 marks

Q10 Stack

5 marks

Point out the error?

PEEK(stack);

return stack[top+1]

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 return stack[top]
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_192211868
RB_6a8bc928eb1d70.40171315\main.c
:1:1: error: expected identifier
or '(' before 'return'
1 | return stack[top]
```

▶ Run

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack Notations - Postfix Expression

5 marks

✓

Q2

Circular Queue

5 marks

✓

Q3

Binary Search

5 marks

✓

Q4

Queue

5 marks

✓

Q5

Queue

5 marks

✓

Q6

Doubly Linked List

5 marks

✓

Q7

Stack

5 marks

✓

Q2 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {  
    if (front == -1) return -1;  
    int data = queue[front];  
    front = (front + 1) % MAX;  
    return data;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int dequeue() {  
2     if (front == -1)  
3         return -1;  
4  
5         int data = queue[front];  
6  
7         if (front == rear){  
8             front = rear - 1;  
9         } else {  
10            front = (front + 1) % MAX;  
11        }  
12        return data;  
}
```

Test Input

stdin input (optional)

Output

```
Compilation Error:  
C:\Windows\TEMP\sandbox_192211868  
RB_6a8bc58481f216.59208668\main.c  
:2:1: error: expected declaration  
specifiers or '...' before 'if'
```